Ch 6 Review Questions Answers

1. A structure that allows repeated execution of a block of statements is a _____.

    a. cycle

    b. ring

    c. loop

    d. band

    Answer: c

    Feedback: A loop is a repeated execution of a block of statement. Although "cycle," "ring," and "band" have similar meanings in English, those terms are not used in programming.

2. A loop that never ends is a(n) _____ loop.

    a. infinite

    b. iterative

    c. structured

    d. illegal

    Answer: a

    Feedback: A loop that never ends is infinite. All loops are iterative whether they are infinite or not, and all well-written loops follow the requirements of structured logic. Infinite loops are not illegal, but they almost always represent a flaw in program logic.

3. To construct a loop that works correctly, you should initialize a loop control _____.

    a. condition

    b. constant

    c. structure

    d. variable

    Answer: d

    Feedback: A loop control variable is initialized before entering a loop.

4. What is the output of the following code?

```
b = 3;

while(b < 6)

   System.out.print(b + " ");
```

    a. 3

b. 3 4 5

c. 3 4 5 6

d. 3 3 3 3 3 3 …

Answer: d

Feedback: The loops displays 3 continuously because the value of b is never altered within the loop.

5. What is the output of the following code?

```
b = 3;

while(b < 6)

{

   System.out.print(b + " ");

   b = b + 1;

}
```

a. 3

b. 3 4 5

c. 3 4 5 6

d. 3 3 3 3 3 3…

Answer: b

Feedback: The loop displays 3 Then b is increased, and the loop displays 4. The value of b is increased, and the loop displays 5. After b is increased to 6, the loop body is not entered again.

6. What is the output of the following code?

```
e = 1;

while(e < 4);

   System.out.print(e + " ");
```

a. nothing

b. 1

c. 1 1 1 1 1…

d. 4 4 4 4 4 4…

Answer: a

Feedback: The loop that starts `while(e < 4)` has an empty body because of the semicolon that follows. So, `e` is continually compared to 4, and the program never progresses past that comparison.

7. If `total = 100` and `amt = 200`, then after the statement `total += amt`, _____.

   a. `total` is equal to 200

   b. `total` is equal to 300

   c. `amt` is equal to 100

   d. `amt` is equal to 300

   Answer: b

   Feedback: The statement `total += amt` adds `amt` to `total`, so `total` becomes 300. The value of `amt` does not change.

8. The prefix ++ is a _____ operator.

   a. unary

   b. binary

   c. tertiary

   d. postfix

   Answer: a

   Feedback: The prefix ++ is a unary operator, meaning it takes one operand.

9. If `g = 5`, then after `h = ++g`, the value of `h` is _____.

   a. 4

   b. 5

   c. 6

   d. 7

   Answer: c

   Feedback: If `g` is 5, then `++g` is 6. If the result is assigned to `h`, then `h` is also 6.

10. If `m = 9`, then after `n = m++`, the value of `m` is _____.

    a. 8

    b. 9

    c. 10

    d. 11

Answer: c

Feedback: The value of `m++` is 9, so 9 is assigned to `n`, but when the statement is complete, `m` is incremented, making it 10.

11. If `m = 9`, then after `n = m++`, the value of `n` is _____.

    a. 8

    b. 9

    c. 10

    d. 11

    Answer: b

    Feedback: The value of `m++` is 9, so 9 is assigned to `n`.

12. If `j = 5` and `k = 6`, then the value of `j++ == k` is _____.

    a. 5

    b. 6

    c. `true`

    d. `false`

    Answer: d

    Feedback: The value of `j++` is 5, and `5 == 6` is `false`. After the statement, `j` becomes 6.

13. You must always include _____ in a `for` loop's parentheses.

    a. two semicolons

    b. three semicolons

    c. two commas

    d. three commas

    Answer: a

    Feedback: A `for` loop's parentheses contain two semicolons that separate the three parts. A `for` loop might have any number of commas within each part.

14. What does the following statement output?

```
for(a = 0; a < 5; ++a)
    System.out.print(a + " ");
```

    a. 0 0 0 0 0

    b. 0 1 2 3 4

c. 0 1 2 3 4 5

d. nothing

Answer: b

Feedback: The loop is entered when the value of `a` is 0, 1, 2, 3, and 4.

15. What does the following statement output?

```
for(b = 1; b > 3; ++b)
   System.out.print(b + " ");
```

a. 1 1 1

b. 1 2 3

c. 1 2 3 4

d. nothing

Answer: d

Feedback: After `b` is initialized to 1, it is compared to 3. The result is `false`, so the loop body is never entered.

16. What does the following statement output?

```
for(f = 1, g = 4; f < g; ++f, --g)
   System.out.print(f + " " + g + " ");
```

a. 1 4 2 5 3 6 4 7…

b. 1 4 2 3 3 2

c. 1 4 2 3

d. nothing

Answer: c

Feedback: When `f` is 1 and `g` is 4, the loop is entered, and 1 and 4 display. Then `f` becomes 2, and `g` becomes 3. The loop is entered again, and 2 and 3 are displayed. Then `f` becomes 3, and `g` becomes 2, so `f < g` is `false`, and the loop is not entered a third time.

17. The loop that performs its conditional check at the bottom of the loop is a _____

loop.

a. `while`

b. `do…while`

c. `for`

d. `for…while`

Answer: b

Feedback: The `do…while` loop performs its check at the bottom after each execution of the loop body.

18. What does the following program segment output?

```
d = 0;

do

{

   System.out.print(d + " ");

   d++;

} while (d < 2);
```

a. 0

b. 0 1

c. 0 1 2

d. nothing

Answer: b

Feedback: The loop is entered, and `0` is displayed. Then `d` becomes `1`, and because it is less than `2`, the loop body executes again, and `1` is displayed. The `d` becomes `2`, so `d` is not less than `2`, so the loop is not entered a third time.

19. What does the following program segment output?

```
for(f = 0; f < 3; ++f)

   for(g = 0; g < 2; ++g)

      System.out.print(f + " " + g + " ");
```

a. 0 0 0 1 1 0 1 1 2 0 2 1

b. 0 1 0 2 0 3 1 1 1 2 1 3

c. 0 1 0 2 1 1 1 2

d. 0 0 0 1 0 2 1 0 1 1 1 2 2 0 2 1 2 2

Answer: a

Feedback: The variable `f` is `0`, then `g` is `1` and `2`, so `0 1 0 2` is displayed. Then `f` is `1` and `g` again becomes `0` and `1`, so `1 0 1 1` is displayed. Finally, `f` becomes `2` and `g` becomes `0` and `1`, so `2 0 2 1` is displayed.

20. What does the following program segment output?

```
for(m = 0; m < 4; ++m);
```

```
for(n = 0; n < 2; ++n);

   System.out.print(m + " " + n + " ");
```

a. 0 0 0 1 1 0 1 1 2 0 2 1 3 0 3 1

b. 0 1 0 2 1 1 1 2 2 1 2 2

c. 4 2

d. 3 1

Answer: c

Feedback: The first `for` loop has an empty body, so `m` becomes, in turn, 0, 1, 2, 3, and 4. The next `for` loop also has an empty body, so `n` becomes, in turn, 0, 1, and 2. The final `print()` statement executes just once, and it displays 4 and 2.